Internet — Full Deep Explanation

🔹 What is the Internet?

The Internet is a massive global system of interconnected networks consisting of:

- Millions of routers
- Data centers
- Servers
- Fiber-optic cables
- Wireless links
- ISPs (Internet Service Providers)

It uses **packet switching**, where data is divided into small packets and routed independently.

🔹 Internet Hierarchy

Tier-1 ISPs

- Global backbone providers
- Own international fiber routes
- Examples: AT&T, Level3, Tata Communications

Tier-2 ISPs

- Regional providers
- Buy bandwidth from Tier-1
- Examples: National ISPs

Tier-3 ISPs

- Local broadband ISPs
- They deliver internet to homes/offices

🔹 **Key Components of the Internet**

- **DNS** → Converts domain names to IP addresses
- **BGP (Border Gateway Protocol)** → Decides how packets travel globally
- **CDN (Cloudflare, Akamai)** → Deliver content fast globally
- **IXP (Internet Exchange Point)** → ISPs interconnect here
- **Undersea Fiber Cables** → Backbone of the Internet

ARP Protocol — Deep Explanation

ARP (Address Resolution Protocol) maps an **IP address → MAC address** inside a LAN.

Why ARP?

Because switches deliver frames using **MAC**, not IP.

ARP Message Types

1. **ARP Request (Broadcast)**
   → Who has 192.168.1.10?
2. **ARP Reply (Unicast)**
   → I am 192.168.1.10, MAC = AA:BB:CC:DD:EE:FF

ARP Cache Example

```
192.168.0.1   00:11:22:33:44:55   dynamic
192.168.0.10  FC:FB:FB:AA:BB:CC   static
```

ARP Attacks

- ARP Spoofing / Poisoning
- Man-in-the-Middle
- Used in Kali Linux: Ettercap, Bettercap, arpspoof

ICMP Protocol — Deep Explanation

ICMP = Internet Control Message Protocol
Used for **error reporting and diagnostics**.

ICMP Works at:

- **Network Layer (Layer 3)**

Common ICMP Types

Type	Description
0	Echo Reply (Ping Reply)
8	Echo Request
3	Destination Unreachable
11	Time Exceeded
5	Redirect Message

ICMP Uses:

- Ping
- Traceroute
- Error handling
- Network troubleshooting

Security Note:

Many firewalls block ICMP to prevent:

- Ping Flood
- Smurf Attack

IGMP Protocol — Deep Explanation

IGMP (Internet Group Management Protocol) manages **Multicast group memberships**.

Why Multicast?

Efficient delivery of:

- IPTV
- Video streaming
- Online seminars
- Live gaming sessions

Multicast IP Range:
224.0.0.0 – 239.255.255.255

IGMP Versions:

- **IGMPv1** – Basic join
- **IGMPv2** – Leave messages added
- **IGMPv3** – Source-specific multicast (more secure)

QUIC Protocol — Deep Explanation
QUIC = Quick UDP Internet Connections
Created by Google.

Used by **HTTP/3**.

QUIC Features:

1. **Zero-RTT connection** (ultra-fast)
2. **Built-in encryption (TLS 1.3)**
3. **Works over UDP, but reliable**
4. **No Head-of-Line Blocking**
5. **Connection migration** (WiFi → Mobile Data without disconnect)

Who Uses QUIC?

- Google
- YouTube
- Facebook
- Cloudflare
- All HTTP/3 websites

Ping Protocol — Deep Explanation
Ping is not a protocol itself.
It uses **ICMP Echo Request/Reply** messages.

Ping Measures:

- Connection availability
- Round Trip Time (RTT)
- Packet loss
- Jitter
- DNS resolution

Example:

64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=32.1 ms

MAC Address — Deep Explanation

MAC = Media Access Control Address
A **48-bit**, globally unique identifier for a network card.

MAC Format:

AA:BB:CC:11:22:33

Structure:

- First 24 bits: **OUI (Manufacturer code)**
- Last 24 bits: **Device specific**

MAC Spoofing Example (Kali Linux)

```
ifconfig wlan0 down
macchanger -r wlan0
ifconfig wlan0 up
```

Network Devices — Deep Technical Explanation

Hub

- Layer 1 device
- Broadcasts all data
- No intelligence

- No MAC table
- Single collision domain
- Uses CSMA/CD

Hub = OLD + INSECURE technology.

Switch

Layer 2 device (sometimes Layer 3).

Switch Features:

- MAC Address Table
- STP (Spanning Tree Protocol)
- VLAN support
- Inter-VLAN routing (Layer 3 switch)
- QoS
- DHCP Snooping
- ARP Inspection
- Trunk links (802.1Q)
- Port Security

Switch = Smart + Secure + Fast.

Router

Layer 3 device.

Router Functions:

- IP Routing
- NAT/PAT
- DHCP Server
- Firewall
- VPN Tunneling
- Interconnecting networks
- Access Control Lists (ACL)

Routing Protocols:

- **OSPF, RIP, EIGRP** → IGP
- **BGP** → Global Internet routing

Bridge

- Older form of switch
- Layer 2
- Divides collision domains
- Slow, replaced by modern switches

Repeater

- Layer 1
- Regenerates weak signals
- Used in fiber and WiFi boosters

Gateway

Converts **one protocol format → another protocol format**.

Examples:

- VoIP Gateway → Converts SIP to PSTN
- Payment Gateway → Handles financial transactions
- IoT Gateway → Converts industrial protocols

A router can act as a gateway, but a gateway is more general.

Access Point

Creates wireless networks (WiFi).

Types:

- Standalone AP
- Controller-based AP
- Mesh AP

AP Features:

- WPA2/WPA3 security
- WiFi 5/6/6E support
- Multi-SSID
- Channel selection
- Band steering
- Load balancing

Modem

Modulator + Demodulator.

Converts ISP signal into digital data for routers.

Types:

- DSL Modem
- Cable Modem
- Fiber ONT (GPON/EPON)

Synchronous & Asynchronous Data Transmission

🔹 **Synchronous Transmission**

Data is sent **with a common clock signal**.
Sender and receiver stay synchronized.

Features:

- Continuous stream
- High speed
- Requires clock sync
- Used in: LAN, USB, DDR RAM, High-speed links

Diagram:

Clock: | | | | | | | |

Data : 1 0 1 1 0 0 1

🔹 Asynchronous Transmission

Data is sent **character-by-character** without a clock signal.
Start/Stop bits indicate beginning and end.

Features:

- Slower
- Simple & low cost
- No clock sync needed
- Used in: Keyboard, Mouse, Serial Port (RS-232)

Diagram:

Start | Data Bits | Stop

0 10101 1

🔹 Difference Table

Feature	Synchronous	Asynchronous
Clock	Required	Not required
Speed	High	Low
Cost	Higher	Lower
Data Flow	Continuous	Byte-by-byte
Examples	DDR, USB, LAN	Keyboard, Serial port

Simplex, Half Duplex, Full Duplex

🔹 Simplex Mode

One-way communication.

Examples:

- TV broadcast
- Keyboard → PC (old systems)

Diagram:

A → B

🔹 **Half Duplex Mode**

Two-way communication, but **one direction at a time**.

Examples:

- Walkie-talkie
- Older wireless systems

Diagram:

A ↔ B (one at a time)

🔹 **Full Duplex Mode**

Two-way communication **simultaneously**.

Examples:

- Telephone
- Modern Ethernet
- Switch communication

Diagram:

A ⇄ B (simultaneous)

🔹 **Difference Table**

Mode	Direction	Example
Simplex	One-way	TV
Half Duplex	Both ways (not same time)	Walkie-talkie
Full Duplex	Both ways (same time)	Phone call

IP Addressing (IPv4 & IPv6)

🔹 **IPv4 (32-bit)**

- Length: 32 bits
- Format: Dotted decimal
- Example: **192.168.1.1**
- Total: **4.3 billion addresses**

IPv4 Classes:

Class	Range	Usage
A	1–126	Large networks
B	128–191	Medium networks
C	192–223	Small networks
D	224–239	Multicast
E	240–255	Experimental

🔹 **IPv6 (128-bit)**

- Length: 128 bits
- Format: Hexadecimal
- Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334
- Total: **340 undecillion addresses**

IPv6 Features:

- No NAT needed
- Auto-configuration (SLAAC)
- Built-in IPsec
- No broadcast (only multicast/anycast)
- Faster routing

Subnetting — Full Explanation

Subnetting = dividing a large network into **smaller sub-networks**.

Why Subnet?

- Better management
- Reduce broadcast traffic
- Improve security
- Efficient IP usage

Example:

Given:
192.168.1.0/24 → 256 addresses

Divide into 2 subnets:

1. 192.168.1.0/25
2. 192.168.1.128/25

Each subnet:

- 128 IP addresses
- 126 usable
- 1 network + 1 broadcast

Supernetting — Full Explanation

Supernetting = combining multiple small networks into one large block.
Also called **Route Aggregation**.

Used by ISPs for global routing.

Example:

Combine:

- 192.168.0.0/24
- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

→ Supernet = **192.168.0.0/22**

CIDR Notation — Full Explanation

CIDR = Classless Inter-Domain Routing

Format: **IP/prefix**

Examples:

- 10.0.0.0/8
- 192.168.1.0/24
- 172.16.0.0/12

Prefix length = network portion.

Common CIDR Blocks:

CIDR	Total IPs	Usable
/8	16,777,216	huge
/16	65,536	large
/24	256	small
/30	4	point-to-point
/32	1	host route

OSI Model — Full Explanation

OSI = Open Systems Interconnection
7 layers.

Layer	Name	Function	Examples
7	Application	Services to user	HTTP, DNS
6	Presentation	Encryption, formatting	SSL/TLS
5	Session	Dialog control	Session IDs
4	Transport	End-to-end delivery	TCP, UDP
3	Network	Routing	IP, ICMP
2	Data Link	MAC, frames	Switch, ARP
1	Physical	Bits, signals	Cable, Hub

TCP/IP Model — Full Explanation

4 layers only.

TCP/IP Layer	OSI Equivalent	Protocols
Application	5,6,7	HTTP, FTP, DNS, SMTP
Transport	4	TCP, UDP
Internet	3	IP, ICMP, ARP
Network Access	1,2	Ethernet, WiFi

Well-Known Ports and Their Functions

These are standard ports assigned to specific network services.

Most Important Ports (Table)

Port	Protocol	Service	Function
20/21	TCP	FTP	File transfer
22	TCP	SSH	Secure remote login
23	TCP	Telnet	Unsecured remote login
25	TCP	SMTP	Sending emails
53	UDP/TCP	DNS	Domain → IP resolution
67/68	UDP	DHCP	Auto IP assignment
69	UDP	TFTP	Simple file transfer
80	TCP	HTTP	Web browsing
110	TCP	POP3	Receive emails
123	UDP	NTP	Time synchronization
143	TCP	IMAP	Email retrieval
161/162	UDP	SNMP	Network monitoring
389	TCP/UDP	LDAP	Directory services
443	TCP	HTTPS	Secure web browsing
445	TCP	SMB	Windows file sharing
514	UDP	Syslog	Logging
5060	UDP/TCP	SIP	VoIP signaling
8080	TCP	HTTP Proxy	Web proxy

TCP & UDP Protocols (When, Where & How They Send Data)

🔹 **TCP (Transmission Control Protocol)**

✔️ **Where is TCP used?**

Anywhere **reliable**, **ordered**, **error-free** communication is required:

- HTTPS (secure browsing)
- Email
- File Transfer (FTP)
- SSH remote login
- Database communication

✔️ **How TCP sends data?**

- Connection-oriented
- Uses **3-way handshake**
- Guarantees ordered data
- Retransmits lost packets
- Provides flow control & congestion control

TCP ensures:
➡️ Data is delivered
➡️ Data is in the correct order
➡️ No data corruption

🔹 **UDP (User Datagram Protocol)**

✔️ **Where is UDP used?**

Where **speed** is more important than reliability:

- Video streaming
- Online gaming
- VoIP calls
- DNS queries
- Live broadcasts

✔️ How UDP sends data?

- Connectionless
- No handshake
- No delivery guarantee
- Fast and lightweight

UDP ensures:
➡️ Very low latency
➡️ Fast transmission
➡️ Allows broadcast & multicast

🔹 Difference Table

Feature	TCP	UDP
Type	Connection-oriented	Connectionless
Speed	Slow	Fast
Reliability	High	Low
Ordering	Guaranteed	Not guaranteed
Use cases	Web, email	Gaming, live streaming

TCP 3-Way Handshake (Connection Establishment)

Used to establish a TCP connection.

Client ---- SYN ---> Server

Client <--- SYN/ACK --- Server

Client ---- ACK ---> Server

Step-by-step:

1. **SYN** → Client: "I want to connect."
2. **SYN/ACK** → Server: "I accept."
3. **ACK** → Client: "Okay, connection ready."

Connection established successfully.

TCP 4-Way Handshake (Connection Termination)

Used when closing a TCP connection.

Client ---- FIN ---> Server

Client <--- ACK ---- Server

Client <--- FIN ---- Server

Client ---- ACK ---> Server

Step-by-step:

1. Client sends **FIN**
2. Server replies **ACK**
3. Server sends its own **FIN**
4. Client replies **ACK**

Both sides close gracefully.

TCP Flags (SYN, ACK, FIN, RST, PSH, URG)

Flags identify the purpose of a TCP packet.

Flag	Full Form	Meaning
SYN	Synchronize	Start connection
ACK	Acknowledgment	Confirm packet received
FIN	Finish	Close connection
RST	Reset	Force close connection
PSH	Push	Send data immediately (skip buffer)
URG	Urgent	High-priority data

Usage Example:

- SYN = Start handshake
- SYN/ACK = Accept handshake
- FIN = Initiate close
- RST = Emergency termination

HTTP Methods (GET, POST, PUT, PATCH, TRACE, HEAD, OPTIONS)

HTTP methods are commands used by a client to interact with a web server.

HTTP Methods Table

Method	Purpose	Description
GET	Retrieve data	Data visible in URL, safe for read-only
POST	Submit data	Used for login, signup, form submission
PUT	Replace resource	Full update of a record
PATCH	Partial update	Modify only specific fields
DELETE	Remove resource	Delete an item
HEAD	Get headers	Same as GET but without body
OPTIONS	Check supported methods	Used for CORS, debugging
TRACE	Loopback test	Shows request path (usually disabled)

🔹 Short Explanation

GET

- Used to request data
- Visible in URL
- Not used for sensitive info

POST

- Sends data in body
- Secure for login/password
- Used in most forms

PUT

- Replaces entire record

PATCH

- Updates only part of the resource

HEAD

- Checks if resource exists

OPTIONS

- Shows allowed HTTP methods (important for CORS)

TRACE

- Returns the request path (security risk, often turned off)

DNS (Domain Name System)

DNS works as **the phonebook of the internet**.
It converts **domain names → IP addresses**.

Example:

google.com → 142.250.190.78

facebook.com → 157.240.200.35

🔹 **How DNS Works (Steps)**

1. User types domain
2. Browser sends DNS query
3. Resolver checks cache
4. Goes to **Root Server**
5. Root → **TLD Server** (.com, .net, etc.)
6. TLD → **Authoritative DNS**
7. Authoritative DNS returns IP
8. Browser connects to the server

🔹 **DNS Record Types**

Record	Purpose
A	IPv4 Address
AAAA	IPv6 Address
CNAME	Alias name
MX	Mail server
TXT	Verification, SPF/DKIM
NS	Name server
PTR	Reverse DNS

DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns:

- IP Address
- Subnet Mask
- Gateway
- DNS Server
- Lease Time

🔹 **DHCP Process (DORA)**

1. **Discover**
2. **Offer**
3. **Request**
4. **Acknowledge**

FTP, TFTP, SFTP, SCP

🔹 **FTP (File Transfer Protocol)**

- Port: **20/21**
- Not encrypted (unsafe)
- Upload/download files

🔹 **TFTP (Trivial FTP)**

- Port: **69**
- No authentication
- Very fast
- Used for router booting, PXE boot

🔹 **SFTP (SSH File Transfer Protocol)**

- Port: **22**
- Secure (SSH-based)
- Secure file uploads/downloads

🔹 **SCP (Secure Copy Protocol)**

- Port: **22**
- Very fast secure copying over SSH

SSH & Telnet

🔹 **Telnet**

- Port: **23**
- No encryption
- Sends data as plain text
- Not recommended

🔹 **SSH (Secure Shell)**

- Port: **22**
- Fully encrypted
- Used for safe remote login
- Used by admins, servers, hackers (pentesting)

SMTP, POP3, IMAP

🔹 **SMTP (Simple Mail Transfer Protocol)**

- Port: **25**
- Used to **send** emails

🔹 **POP3 (Post Office Protocol v3)**

- Port: **110**
- Downloads emails to device
- Often deletes from server

🔹 **IMAP (Internet Message Access Protocol)**

- Port: **143**
- Sync emails across devices
- Keeps mail on server

SNMP (Simple Network Management Protocol)

Used for **network monitoring and management**.

SNMP Monitors:

- Router/Switch health
- Traffic usage
- Errors
- CPU load
- Interface up/down

Ports:

- **161** – Queries
- **162** – Alerts (Traps)

NTP (Network Time Protocol)

- Port: **123 (UDP)**
- Synchronizes time across devices
- Used in servers, routers, banks, security systems

Example:
Google NTP → time.google.com

SMB, LDAP, RDP

🔹 **SMB (Server Message Block)**

- Port: **445**
- Windows File Sharing
- Printer sharing
- AD communication
- Vulnerable to: EternalBlue, WannaCry

🔹 **LDAP (Lightweight Directory Access Protocol)**

- Port: **389**
- Used for authentication & directory lookups
- Used in Active Directory (AD)

🔹 **RDP (Remote Desktop Protocol)**

- Port: **3389**
- Used for Windows remote access
- Full graphical remote control

NAT & PAT (Network and Port Address Translation)

🔹 **NAT**

Translates private IP → public IP.

Example:

192.168.1.5 → 203.0.113.10

🔹 **PAT (NAT Overload)**

Multiple private devices share ONE public IP using different port numbers.

Example:

192.168.1.10:5050 → 203.0.113.10:15501

192.168.1.11:5050 → 203.0.113.10:15502

Used in home routers.

Firewalls & CDN

🔹 **Firewall**

Controls and filters traffic based on:

- IP
- Port
- Protocol
- Application

- URL
- Content
- Behavior

Types:

- Packet-filtering firewall
- Stateful firewall
- Proxy firewall
- NGFW (Next-Gen Firewall)

🔹 CDN (Content Delivery Network)

CDN delivers content fast across the world using:

- Global caching
- Load balancing
- DDoS protection
- Smart routing

Examples:
Cloudflare, Akamai, Fastly

VPN Types (SSL, IPsec, PPTP, L2TP)

🔹 SSL VPN

- Runs over HTTPS (443)
- Easy to use
- Perfect for remote workers

🔹 IPsec VPN

- Most secure
- Works at **Network Layer**
- Used for site-to-site VPN
- Strong encryption

🔹 **PPTP**

- Very old
- Weak encryption
- Not recommended

🔹 **L2TP/IPsec**

- Combines L2TP + IPsec
- More secure + stable
- Widely used in enterprise environments

TLS (Transport Layer Security)

TLS = Security layer for HTTPS.

TLS Provides:

- **Encryption** (data secure)
- **Authentication** (server verified)
- **Integrity** (data not changed)
- **Protection from MITM attacks**

Versions:

- TLS 1.0 → Deprecated
- TLS 1.2 → Standard
- TLS 1.3 → Fastest, Most Secure

Netstat & Traceroute

🔹 **Netstat**

Netstat = **Network Statistics**
Used to view all network connections in a system.

Netstat Shows:

- Active TCP/UDP connections
- Listening ports
- Established connections
- Routing table
- Interfaces

Common Command:

netstat -tulnp

Flag	Meaning
-t	TCP
-u	UDP
-l	Listening
-n	Numeric IP/ports
-p	Process ID

🔹 **Traceroute**

Traceroute shows the **path (hops)** a packet takes to reach its destination.

How it works?

Traceroute increases TTL step-by-step:

- TTL=1 → 1st router responds
- TTL=2 → 2nd router responds

Command:

traceroute google.com

Encryption Protocols (WEP, WPA, WPA2, WPA3)

🔹 **WEP**

- Very weak
- Uses RC4
- Easily cracked
- Deprecated

🔹 **WPA**

- Uses TKIP
- More secure than WEP
- Still weak (WPS attack possible)

🔹 **WPA2**

- Most common
- Uses AES-CCMP
- Strong security

🔹 **WPA3**

- Uses SAE (Dragonfly handshake)
- Resists KRACK
- Strongest WiFi security

Wireless Authentication (PSK, Enterprise)

🔹 **PSK (Pre-Shared Key)**

- Shared WiFi password
- Used in home networks
- Everyone uses same key

🔹 **Enterprise (802.1X + RADIUS)**

- Username & password login
- Uses RADIUS server
- Used in organizations

Comparison:

Feature	PSK	Enterprise
Key	Shared	Per user
Security	Medium	High
Use	Home	Corporate

Wireless Attacks (Deauth, Evil Twin, KRACK)

🔹 **1) Deauthentication Attack**

- Sends fake deauth frames
- Disconnects users from WiFi
- Used to capture WPA handshake
- Tool: aireplay-ng

🔹 **2) Evil Twin**

- Creates a fake Access Point
- Victim connects to fake AP
- Attacker performs MITM
- Used to steal credentials

🔹 **3) KRACK Attack**

- Weakness in WPA2 handshake
- Attacker decrypts traffic
- Works using key reinstallation attacks

ARP Spoofing / ARP Poisoning

Attacker sends fake ARP replies to poison the ARP table.

Goal:

Redirect victim's traffic through attacker.

Results:

- MITM
- Sniffing passwords
- DNS spoofing
- Session hijacking

Tools:

- Ettercap
- Bettercap
- arpspoof

DNS Spoofing

Fake DNS responses to redirect a victim to a malicious IP.

Example:

facebook.com → 192.168.1.50 (attacker)

Used for:

- Phishing
- Credential theft
- MITM

IP Spoofing

Attacker sends packets with a **fake source IP address**.

Used for:

- Hiding identity
- SYN flood attacks
- Reflection/Amplification attacks
- Bypassing IP-based filters

MAC Flooding

Attacker floods the switch with random MAC addresses.

Effect:

- Switch MAC table becomes full
- Switch enters **hub mode**
- Attacker can sniff all traffic

Tool:

macof (Dsniff package)

DHCP Starvation

Attacker sends many fake DHCP discover requests.

Result:

- DHCP server runs out of IP addresses
- Legit users cannot get IP
- Attacker sets up rogue DHCP server

Tool:

Yersinia

SYN Flood / DoS / DDoS

🔹 **SYN Flood**

Attacker sends huge SYN packets → never completes handshake.

Result:

- Server resources exhausted
- Half-open table full
- Server crashes or slows down

🔹 **DoS**

One attacker → floods a target.

🔹 **DDoS**

Many distributed attackers (botnets) flood a target.

Botnets examples:

- Mirai
- Zeus

MITM (Man-in-the-Middle)

Attacker sits between two communicating parties.

Techniques:

- ARP Spoofing
- DNS Spoofing
- Evil Twin
- SSL Stripping
- HTTPS Downgrade

Attacker can:

- Read data
- Modify data
- Steal credentials
- Inject malicious scripts

Sniffing & Eavesdropping

🔹 **Sniffing**

Sniffing means **capturing network traffic** to read data packets.

Types:

✔️ **Passive Sniffing**

- Attacker only listens
- Does not modify traffic
- Works in hub/broadcast networks

✔️ **Active Sniffing**

- Attacker manipulates traffic
- Works in switched networks
- Techniques:
    - ARP Poisoning
    - MAC Flooding
    - DHCP Spoofing

Tools:

- Wireshark
- tcpdump
- Bettercap
- Ettercap

🔹 Eavesdropping

Secretly listening to communication.

In network security:

- Happens when traffic is unencrypted
- Attacker steals usernames, passwords, cookies

Eavesdropping = covert sniffing.

Replay Attacks

Attacker **captures legitimate packets** and **re-sends them later**.

Examples:

- Replaying login requests
- Repeating financial transaction packets
- Reusing authentication tokens

Prevention:

- Nonce
- Time-stamps
- Session tokens
- TLS/HTTPS

Session Hijacking

Attacker takes over an **active user session**.

Methods:

- Cookie hijacking
- Session ID prediction
- MITM-based session takeover

Steps:

1. Capture session cookie
2. Use cookie to log in as victim

Tools:

- Burp Suite
- Bettercap
- Wireshark

ICMP Flood, Ping of Death, Smurf Attack

🔹 **ICMP Flood**

Attacker sends massive ICMP Echo (ping) requests.

Result:

- Target CPU overload
- Network congestion
- DoS condition

🔹 **Ping of Death**

- Sends oversized ICMP packets
- Causes crash or overflow
- Old systems vulnerable

🔹 **Smurf Attack**

- Attacker spoofs victim's IP
- Sends ICMP to broadcast address
- Whole network replies to victim
- Victim gets overwhelmed

VLAN Hopping

Attacker jumps from one VLAN to another VLAN.

Techniques:

1. **Switch Spoofing**
    - Attacker pretends to be a trunk port
2. **Double Tagging**
    - Uses 2 VLAN tags to bypass VLAN boundaries

Prevention:

- Disable DTP
- Set all ports to access mode
- Use VLAN pruning

Rogue DHCP & Evil Twin

🔹 **Rogue DHCP**

Attacker creates a fake DHCP server.

Victim receives:

- Wrong gateway
- Wrong DNS
- Wrong IP

Attacker can then:

- Redirect traffic
- Perform MITM
- Conduct DNS spoofing

🔹 **Evil Twin**

A fake WiFi Access Point identical to the legitimate one.

Uses:

- MITM
- Credential theft
- Packet sniffing

Tools:

- Airbase-ng
- hostapd-wpe

Routing Attacks

Routing attacks target routing tables or routing protocols.

Types:

✔️ **Route Injection**

Attacker injects fake routes.

✔️ **Blackhole Attack**

Attacker absorbs/drops all traffic.

✔️ **Routing Table Poisoning**

Modifies routing entries with malicious information.

Targeted protocols:

- BGP
- OSPF
- RIP

SSL Stripping

Attacker downgrades **HTTPS → HTTP** to steal data.

How it works:

1. Victim sends HTTPS request
2. Attacker intercepts
3. Attacker connects HTTPS to server
4. Victim receives HTTP
5. Attacker reads and modifies everything

Tools:

- sslstrip
- Bettercap
- Burp Suite

Port Scanning Detection & Prevention

Detection:

- IDS/IPS (Snort, Suricata)
- Firewall logs
- Abnormal SYN patterns
- Nmap activity detection

Prevention:

- SYN cookies
- Port knocking
- Disable unnecessary services
- Rate limiting
- Strong firewall rules

Pivoting Through Networks

After compromising one machine, attacker uses it to access **internal networks**.

Techniques:

- SSH pivoting
- SOCKS proxy
- Proxychains
- Metasploit route add
- VPN tunneling

Example:

Attacker → Compromised PC → Internal servers

Traffic Redirection & Tunneling

🔹 **Traffic Redirection**

Attacker redirects victim traffic:

- DNS spoofing
- ARP poisoning
- Route manipulation

🔹 **Tunneling**

Hiding one protocol inside another to bypass firewalls.

Examples:

- SSH tunnel
- HTTP tunnel
- DNS tunnel
- ICMP tunnel

Used for bypassing restrictions.

Reverse Shells & Bind Shells

🔹 **Reverse Shell**

Victim connects back to attacker.

Victim → Attacker listener

Used because:

- Firewalls often block inbound traffic
- But allow outbound traffic

Tools:

- Netcat
- Metasploit
- Python payloads

🔹 **Bind Shell**

Victim opens a port and attacker connects.

Attacker → Victim open port

Harder to bypass firewalls.